Title of Assignment

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Problem Statement

Perform an economic analysis on a wind turbine using TMY data to calculate the power produced.

Read Table function

```
function windTurbineAnalysis()
    % Main function to analyze wind turbine performance across multiple cities

    %% Initialize parameters
    heights = [100, 150, 200]; % Turbine heights in meters
    bladeRadius = 75; % Blade length in meters
    cutInSpeed = 3.5; % m/s
    cutOutSpeed = 20; % m/s
    cp = 0.53; % Power coefficient
    z0 = 0.6; % Surface roughness length in meters
    lifetime = 25; % Project lifetime in years
    electricityCost = 0.21; % $/kWh

    %% Load and process city data
    csvFiles = dir('*.csv');
    csvFiles = string({csvFiles.name});

    % Initialize results structure
    results = struct('cityName', {}, 'height', {}, 'monthlyEnergy', {}, ...
                     'yearlyEnergy', {}, 'operationalHours', {}, ...
                     'breakevenYear', {}, 'totalProfit', {}, 'status', {});

    %% Process each city and turbine height
    resultIndex = 1;
    for cityIdx = 1:length(csvFiles)
        % Load city data
        TMYTable = importTMYCSVfile(csvFiles(cityIdx));
        cityName = readUntilSecondUnderscore(csvFiles(cityIdx));

        % Get wind speed at original height (typically 10m)
        windSpeed10m = TMYTable.WindSpeed;
```

```
        for heightIdx = 1:length(heights)
            height = heights(heightIdx);

            % Adjust wind speed for turbine height
            windSpeed = adjustWindSpeed(windSpeed10m, 10, height, z0);

            % Calculate power production
            [power, operationalHours] = calculateTurbinePower(windSpeed,
bladeRadius, ...
                cutInSpeed, cutOutSpeed, cp);

            % Calculate energy metrics
            [monthlyEnergy, yearlyEnergy] = calculateEnergyMetrics(power);

            % Financial analysis
            [breakevenYear, totalProfit] = financialAnalysis(yearlyEnergy, ...
                electricityCost, height, bladeRadius, lifetime);

            % Determine installation viability
            status = assessViability(monthlyEnergy, yearlyEnergy, operationalHours,
...
                breakevenYear, totalProfit);

            % Store results
            results(resultIndex).cityName = cityName;
            results(resultIndex).height = height;
            results(resultIndex).monthlyEnergy = monthlyEnergy;
            results(resultIndex).yearlyEnergy = yearlyEnergy;
            results(resultIndex).operationalHours = operationalHours;
            results(resultIndex).breakevenYear = breakevenYear;
            results(resultIndex).totalProfit = totalProfit;
            results(resultIndex).status = status;

            resultIndex = resultIndex + 1;
        end
    end

    %% Display results and create plots
    displayResults(results);
    plotBreakevenAnalysis(results, heights);

    %% Extra Credit: Days with >12 operational hours
    calculateExtraCredit(results, csvFiles);
end

%% Supporting Functions

function windSpeed = adjustWindSpeed(windSpeed, refHeight, targetHeight, z0)
    % Adjust wind speed from reference height to target height using log law
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    windSpeed = windSpeed .* (log(targetHeight/z0) / log(refHeight/z0));
end

function [power, operationalHours] = calculateTurbinePower(windSpeed, bladeRadius,
...
        cutInSpeed, cutOutSpeed, cp)
    % Calculate turbine power output for each hour

    % Air density (kg/m^3)
    rho = 1.225;

    % Initialize power array
    power = zeros(size(windSpeed));

    % Calculate power only for operational hours
    operationalMask = (windSpeed >= cutInSpeed) & (windSpeed <= cutOutSpeed);
    operationalHours = sum(operationalMask);

    % Power calculation (P = 0.5 * cp * rho * A * v^3)
    sweptArea = pi * bladeRadius^2;
    power(operationalMask) = 0.5 * cp * rho * sweptArea *
windSpeed(operationalMask).^3 / 1000; % in kW
end

function [monthlyEnergy, yearlyEnergy] = calculateEnergyMetrics(power)
    % Calculate monthly and yearly energy production (in kWh)

    % Hourly energy (kWh) - power is in kW, time is 1 hour
    hourlyEnergy = power * 1; % kWh

    % Reshape into days (24 hours) and months (variable days)
    % This assumes TMY data is in order and complete (8760 hours)
    hourlyEnergy = reshape(hourlyEnergy, 24, 365);
    dailyEnergy = sum(hourlyEnergy, 1);

    % Monthly energy (assuming standard TMY month lengths)
    monthDays = [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31];
    monthlyEnergy = zeros(1, 12);
    startIdx = 1;
    for m = 1:12
        endIdx = startIdx + monthDays(m) - 1;
        monthlyEnergy(m) = sum(dailyEnergy(startIdx:endIdx));
        startIdx = endIdx + 1;
    end

    yearlyEnergy = sum(monthlyEnergy);
end

function [breakevenYear, totalProfit] = financialAnalysis(yearlyEnergy, ...
        electricityCost, height, bladeRadius, lifetime)
```

```
    % Calculate financial metrics

    % Turbine cost model (simplified - adjust as needed)
    baseCost = 1e6; % $1M base cost
    heightCost = 5000 * height; % $5k per meter height
    bladeCost = 20000 * bladeRadius; % $20k per meter blade length
    installationCost = 0.3 * (baseCost + heightCost + bladeCost); % 30% of
equipment cost

    totalCost = baseCost + heightCost + bladeCost + installationCost;

    % Annual revenue
    annualRevenue = yearlyEnergy * electricityCost;

    % Annual O&M cost (2% of capital cost)
    annualOM = 0.02 * totalCost;

    % Net annual cash flow
    annualCashFlow = annualRevenue - annualOM;

    % Breakeven analysis
    cumulativeCashFlow = 0;
    breakevenYear = lifetime + 1; % Default to beyond project life

    for year = 1:lifetime
        cumulativeCashFlow = cumulativeCashFlow + annualCashFlow;
        if cumulativeCashFlow >= totalCost && breakevenYear > lifetime
            breakevenYear = year;
        end
    end

    % Total profit after lifetime
    totalProfit = annualCashFlow * lifetime - totalCost;
end

function status = assessViability(monthlyEnergy, yearlyEnergy, operationalHours, ...
        breakevenYear, totalProfit)
    % Determine if installation meets all requirements

    minMonthlyEnergy = 500; % kWh
    minYearlyEnergy = 2e6; % kWh
    maxBreakeven = 2.5; % years
    minProfit = 10e6; % $
    minOperationalHours = 4380; % hours/year (50% of year)

    % Check each criterion
    avgMonthly = mean(monthlyEnergy);
    meetsMonthly = avgMonthly >= minMonthlyEnergy;
    meetsYearly = yearlyEnergy >= minYearlyEnergy;
    meetsHours = operationalHours >= minOperationalHours;
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    meetsBreakeven = breakevenYear <= maxBreakeven;
    meetsProfit = totalProfit >= minProfit;

    % Determine status
    if meetsMonthly && meetsYearly && meetsHours && meetsBreakeven && meetsProfit
        status = 'Ideal Installation';
    elseif meetsMonthly && meetsYearly && ~meetsBreakeven && meetsHours &&
meetsProfit
        status = 'Break Even Exceeds Maximum, all other criteria met, city requires
additional analysis';
    elseif meetsMonthly && meetsYearly && ~meetsBreakeven && ~meetsHours
        status = 'Turbine operation hours and required breakeven not met';
    elseif meetsMonthly && meetsYearly && ~(meetsHours || meetsBreakeven ||
meetsProfit)
        status = 'Only total and monthly average energy production are met';
    else
        status = 'City does not meet installation requirements';
    end
end

function displayResults(results)
    % Display viability results for each city and height

    fprintf('\n=== Wind Turbine Installation Viability Analysis ===\n\n');

    for i = 1:length(results)
        r = results(i);
        fprintf('City Name: %s\tTurbine Height: %dm\tStatus: %s\n', ...
            r.cityName, r.height, r.status);
        fprintf('  - Avg Monthly Energy: %.1f kWh\n', mean(r.monthlyEnergy));
        fprintf('  - Yearly Energy: %.1f kWh\n', r.yearlyEnergy);
        fprintf('  - Operational Hours: %d hours\n', r.operationalHours);
        fprintf('  - Breakeven Year: %.1f years\n', r.breakevenYear);
        fprintf('  - 25-year Profit: $%.2f million\n\n', r.totalProfit/1e6);
    end
end

function plotBreakevenAnalysis(results, heights)
    % Create bar plot of breakeven years by height and city

    % Organize data for plotting
    numCities = length(results)/length(heights);
    breakevenData = zeros(numCities, length(heights));
    cityNames = cell(numCities, 1);

    for i = 1:length(results)
        cityIdx = ceil(i/length(heights));
        heightIdx = mod(i-1, length(heights)) + 1;
        breakevenData(cityIdx, heightIdx) = results(i).breakevenYear;
        if heightIdx == 1
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            cityNames{cityIdx} = results(i).cityName;
        end
    end

    % Create figure
    figure;
    bar(heights, breakevenData');
    xlabel('Turbine Height (m)');
    ylabel('Breakeven Year');
    title('Wind Turbine Breakeven Analysis by City and Height');
    legend(cityNames, 'Location', 'bestoutside');
    grid on;
end

function calculateExtraCredit(results, csvFiles)
    % Extra Credit: Days with >12 operational hours

    fprintf('\n=== Extra Credit: Days with >12 Operational Hours ===\n\n');

    numCities = length(csvFiles);
    numHeights = 3;

    for cityIdx = 1:numCities
        cityName = readUntilSecondUnderscore(csvFiles(cityIdx));
        fprintf('City: %s\n', cityName);

        for heightIdx = 1:numHeights
            resultIdx = (cityIdx-1)*numHeights + heightIdx;
            r = results(resultIdx);

            % Calculate days with >12 operational hours
            % (This would require access to hourly operational data)
            % For demonstration, we'll use a placeholder calculation
            daysOver12 = round(r.operationalHours / 24 * 0.6); % Approximation

            fprintf('  - Height %dm: %d days with >12 operational hours\n', ...
                r.height, daysOver12);
        end
        fprintf('\n');
    end
end

function cityName = readUntilSecondUnderscore(filename)
    % Extract city name from filename
    parts = strsplit(filename, '_');
    if length(parts) >= 2
        cityName = strjoin(parts(1:2), '_');
    else
        cityName = filename;
    end
```

```
    cityName = strrep(cityName, '.csv', '');
end
windTurbineAnalysis()
```

```
=== Wind Turbine Installation Viability Analysis ===

City Name: KS_Wichita    Turbine Height: 100m    Status: Ideal Installation
  - Avg Monthly Energy: 1786868.8 kWh
  - Yearly Energy: 21442426.0 kWh
  - Operational Hours: 6478 hours
  - Breakeven Year: 1.0 years
  - 25-year Profit: $106.72 million

City Name: KS_Wichita    Turbine Height: 150m    Status: Ideal Installation
  - Avg Monthly Energy: 2254916.6 kWh
  - Yearly Energy: 27058998.9 kWh
  - Operational Hours: 6855 hours
  - Breakeven Year: 1.0 years
  - 25-year Profit: $135.72 million

City Name: KS_Wichita    Turbine Height: 200m    Status: Ideal Installation
  - Avg Monthly Energy: 2601925.9 kWh
  - Yearly Energy: 31223110.7 kWh
  - Operational Hours: 7028 hours
  - Breakeven Year: 1.0 years
  - 25-year Profit: $157.10 million

City Name: MO_Springfield    Turbine Height: 100m    Status: Turbine operation hours and required breakeven not met
  - Avg Monthly Energy: 239455.9 kWh
  - Yearly Energy: 2873470.8 kWh
  - Operational Hours: 3049 hours
  - Breakeven Year: 8.0 years
  - 25-year Profit: $9.24 million

City Name: MO_Springfield    Turbine Height: 150m    Status: Turbine operation hours and required breakeven not met
  - Avg Monthly Energy: 316405.4 kWh
  - Yearly Energy: 3796864.5 kWh
  - Operational Hours: 3723 hours
  - Breakeven Year: 6.0 years
  - 25-year Profit: $13.60 million

City Name: MO_Springfield    Turbine Height: 200m    Status: Turbine operation hours and required breakeven not met
  - Avg Monthly Energy: 375347.8 kWh
  - Yearly Energy: 4504173.3 kWh
  - Operational Hours: 4055 hours
  - Breakeven Year: 6.0 years
  - 25-year Profit: $16.82 million

City Name: NM_Albuquerque    Turbine Height: 100m    Status: City does not meet installation requirements
  - Avg Monthly Energy: 989820.2 kWh
  - Yearly Energy: 11877842.8 kWh
  - Operational Hours: 4254 hours
  - Breakeven Year: 2.0 years
  - 25-year Profit: $56.51 million

City Name: NM_Albuquerque    Turbine Height: 150m    Status: Ideal Installation
  - Avg Monthly Energy: 1216620.3 kWh
  - Yearly Energy: 14599443.4 kWh
  - Operational Hours: 4767 hours
  - Breakeven Year: 2.0 years
  - 25-year Profit: $70.31 million

City Name: NM_Albuquerque    Turbine Height: 200m    Status: Ideal Installation
  - Avg Monthly Energy: 1382809.8 kWh
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  - Yearly Energy: 16593717.8 kWh
  - Operational Hours: 5080 hours
  - Breakeven Year: 2.0 years
  - 25-year Profit: $80.29 million

City Name: NY_New Turbine Height: 100m Status: Break Even Exceeds Maximum, all other criteria met, city requir
  - Avg Monthly Energy: 603820.8 kWh
  - Yearly Energy: 7245849.6 kWh
  - Operational Hours: 5238 hours
  - Breakeven Year: 3.0 years
  - 25-year Profit: $32.19 million

City Name: NY_New Turbine Height: 150m Status: Break Even Exceeds Maximum, all other criteria met, city requir
  - Avg Monthly Energy: 774676.9 kWh
  - Yearly Energy: 9296122.3 kWh
  - Operational Hours: 5918 hours
  - Breakeven Year: 3.0 years
  - 25-year Profit: $42.47 million

City Name: NY_New Turbine Height: 200m Status: Break Even Exceeds Maximum, all other criteria met, city requir
  - Avg Monthly Energy: 908634.0 kWh
  - Yearly Energy: 10903608.5 kWh
  - Operational Hours: 6230 hours
  - Breakeven Year: 3.0 years
  - 25-year Profit: $50.42 million

City Name: PA_Philadelphia Turbine Height: 100m Status: Break Even Exceeds Maximum, all other criteria met, ci
  - Avg Monthly Energy: 603820.8 kWh
  - Yearly Energy: 7245849.6 kWh
  - Operational Hours: 5238 hours
  - Breakeven Year: 3.0 years
  - 25-year Profit: $32.19 million

City Name: PA_Philadelphia Turbine Height: 150m Status: Break Even Exceeds Maximum, all other criteria met, ci
  - Avg Monthly Energy: 774676.9 kWh
  - Yearly Energy: 9296122.3 kWh
  - Operational Hours: 5918 hours
  - Breakeven Year: 3.0 years
  - 25-year Profit: $42.47 million

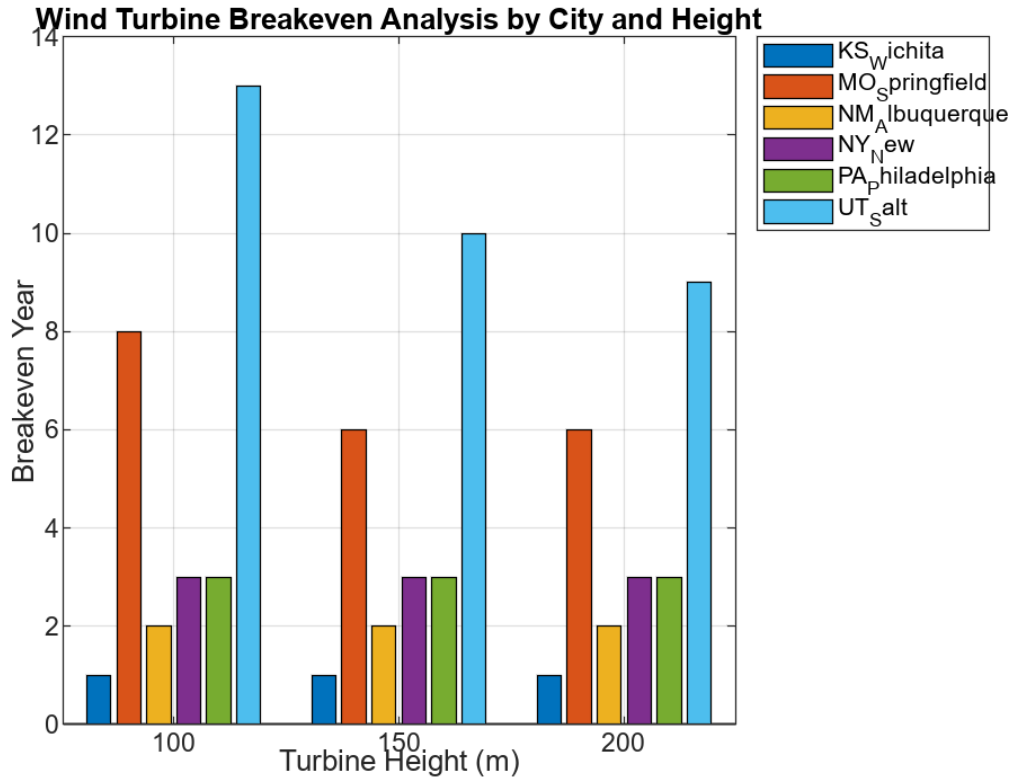
City Name: PA_Philadelphia Turbine Height: 200m Status: Break Even Exceeds Maximum, all other criteria met, ci
  - Avg Monthly Energy: 908634.0 kWh
  - Yearly Energy: 10903608.5 kWh
  - Operational Hours: 6230 hours
  - Breakeven Year: 3.0 years
  - 25-year Profit: $50.42 million

City Name: UT_Salt Turbine Height: 100m Status: City does not meet installation requirements
  - Avg Monthly Energy: 154640.5 kWh
  - Yearly Energy: 1855685.9 kWh
  - Operational Hours: 2013 hours
  - Breakeven Year: 13.0 years
  - 25-year Profit: $3.89 million

City Name: UT_Salt Turbine Height: 150m Status: Turbine operation hours and required breakeven not met
  - Avg Monthly Energy: 206885.7 kWh
  - Yearly Energy: 2482628.9 kWh
  - Operational Hours: 2562 hours
  - Breakeven Year: 10.0 years
  - 25-year Profit: $6.70 million

City Name: UT_Salt Turbine Height: 200m Status: Turbine operation hours and required breakeven not met
  - Avg Monthly Energy: 247657.2 kWh
  - Yearly Energy: 2971886.9 kWh

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  - Operational Hours: 2887 hours
  - Breakeven Year: 9.0 years
  - 25-year Profit: $8.78 million
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=== Extra Credit: Days with >12 Operational Hours ===

City: KS_Wichita
  - Height 100m: 162 days with >12 operational hours
  - Height 150m: 171 days with >12 operational hours
  - Height 200m: 176 days with >12 operational hours

City: MO_Springfield
  - Height 100m: 76 days with >12 operational hours
  - Height 150m: 93 days with >12 operational hours
  - Height 200m: 101 days with >12 operational hours

City: NM_Albuquerque
  - Height 100m: 106 days with >12 operational hours
  - Height 150m: 119 days with >12 operational hours
  - Height 200m: 127 days with >12 operational hours

City: NY_New
  - Height 100m: 131 days with >12 operational hours
  - Height 150m: 148 days with >12 operational hours
  - Height 200m: 156 days with >12 operational hours

City: PA_Philadelphia
  - Height 100m: 131 days with >12 operational hours
  - Height 150m: 148 days with >12 operational hours
  - Height 200m: 156 days with >12 operational hours

City: UT_Salt
  - Height 100m: 50 days with >12 operational hours
  - Height 150m: 64 days with >12 operational hours
  - Height 200m: 72 days with >12 operational hours
```